

LABORATORY RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 2

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1. TITLE

Design and Style a Modern Webpage Using HTML5 and CSS3

2. AIM

To develop a structured and responsive webpage using HTML5 and CSS3 with navigation, content sections, service cards, a contact form, Flexbox layout, and responsive styling.

3. OBJECTIVE

- To understand the use of semantic HTML5 elements.
 - To design a webpage using CSS3 styling properties.
 - To create a header and navigation bar.
 - To divide webpage information into separate sections.
 - To arrange content using CSS Flexbox.
 - To create and style service information cards.
 - To design a simple contact form.
 - To apply hover effects to webpage elements.
 - To make the webpage suitable for smaller screen sizes using media queries.
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4. THEORY

HTML5 provides the basic structure of modern webpages through semantic elements such as header, nav, main, section, article, and footer. These elements make the webpage organized and

easier to understand. CSS3 is used to control the appearance and layout of HTML elements. It can be used to define colors, fonts, spacing, borders, backgrounds, alignment, and other visual properties.

CSS Flexbox provides an efficient way to arrange multiple elements in a row or column. In this webpage, Flexbox is used to arrange the course or service cards.

Form elements such as text fields, email fields, textarea, and buttons are styled using CSS properties. Hover effects provide visual feedback when the user moves the pointer over navigation links or buttons.

A media query is also used to modify the layout for smaller screen sizes. This helps make the webpage more responsive and user-friendly on different devices.

5. ALGORITHM / PROCEDURE

1. Create an HTML5 document named contact.html.
 2. Add the character encoding and viewport settings.
 3. Set the webpage title and connect the CSS stylesheet.
 4. Create a header containing the webpage title and description.
 5. Add navigation links for Home, About, Courses, and Contact.
 6. Create an About section with a description of the learning portal.
 7. Create a Courses section containing three article cards.
 8. Place the article cards inside a Flexbox container.
 9. Create a Contact section containing name, email, message, and submit controls.
 10. Add a footer with copyright information.
 11. Apply CSS rules for colors, spacing, cards, forms, buttons, and navigation.
 12. Add hover effects to interactive elements.
 13. Add a media query to change the layout on smaller screens.
 14. Open the webpage in a browser and verify the output.
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6. PROGRAM

contact.html

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
```

```

<meta name="viewport"
      content="width=device-width, initial-scale=1.0">

<title>Student Learning Portal</title>

<link rel="stylesheet" href="style.css">
</head>

<body>

  <header>
    <h1>Student Learning Portal</h1>
    <p>Explore Courses and Learning Resources</p>
  </header>

  <nav>
    <a href="#home">Home</a>
    <a href="#about">About</a>
    <a href="#courses">Courses</a>
    <a href="#contact">Contact</a>
  </nav>

  <main>

    <section id="about">

      <h2>About Our Portal</h2>

      <p>
        Our learning portal provides students with
        useful courses and resources for improving
        their technical knowledge.
      </p>

    </section>

    <section id="courses">

      <h2>Available Courses</h2>

      <div class="container">

        <article>
          <h3>Web Development</h3>
          <p>
            Learn the fundamentals of HTML,
            CSS and modern webpage development.
          </p>
        </article>

        <article>
          <h3>Python Programming</h3>
          <p>
            Develop programming skills using
            Python and solve basic problems.
          </p>
        </article>

      </div>

    </section>

  </main>

</body>
</html>

```

```

        </p>
    </article>

    <article>
        <h3>Data Science</h3>
        <p>
            Understand data analysis and
            important concepts of data science.
        </p>
    </article>

</div>

</section>

<section id="contact">

    <h2>Get In Touch</h2>

    <form>

        <label for="student">Student Name:</label>

        <input type="text"
            id="student"
            placeholder="Enter student name">

        <label for="email">Email Address:</label>

        <input type="email"
            id="email"
            placeholder="Enter email address">

        <label for="message">Your Message:</label>

        <textarea id="message"
            rows="5"
            placeholder="Type your message">
        </textarea>

        <button type="submit">Send</button>

    </form>

</section>

</main>

<footer>
    <p>&copy; 2026 Student Learning Portal</p>
</footer>

</body>

</html>

```

Style.css

```
* {
    box-sizing: border-box;
    margin: 0;
    padding: 0;
}

body {
    font-family: Arial, sans-serif;
    background-color: #f4f6f8;
    color: #333;
    line-height: 1.6;
}

/* Header */
header {
    background-color: #5e35b1;
    color: white;
    text-align: center;
    padding: 30px;
}

header h1 {
    margin-bottom: 8px;
}

/* Navigation */
nav {
    background-color: #292929;
    padding: 15px;
    text-align: center;
}

nav a {
    color: white;
    text-decoration: none;
    margin: 0 18px;
    font-weight: bold;
}

nav a:hover {
    color: #d1c4e9;
}

/* Main content */
main {
    width: 90%;
    max-width: 1000px;
    margin: 25px auto;
}
```

```

/* Sections */
section {
    background-color: white;
    padding: 25px;
    margin-bottom: 25px;
    border-radius: 10px;
    box-shadow: 0 2px 8px #ccc;
}

section h2 {
    color: #5e35b1;
    margin-bottom: 12px;
}

/* Course cards */
.container {
    display: flex;
    gap: 20px;
    margin-top: 20px;
}

article {
    flex: 1;
    background-color: #ede7f6;
    padding: 20px;
    border-radius: 8px;
    border-left: 5px solid #5e35b1;
}

article h3 {
    color: #4527a0;
    margin-bottom: 8px;
}

/* Form */
form {
    display: flex;
    flex-direction: column;
    margin-top: 15px;
}

label {
    margin-top: 12px;
    margin-bottom: 5px;
    font-weight: bold;
}

input,
textarea {
    width: 100%;
    padding: 11px;
    border: 1px solid #aaa;
    border-radius: 5px;
    font-size: 15px;
}

```

```

textarea {
  resize: vertical;
}

button {
  width: 100px;
  margin-top: 18px;
  padding: 11px;
  background-color: #5e35b1;
  color: white;
  border: none;
  border-radius: 5px;
  cursor: pointer;
}

button:hover {
  background-color: #4527a0;
}

/* Footer */
footer {
  background-color: #292929;
  color: white;
  text-align: center;
  padding: 18px;
  margin-top: 30px;
}

/* Responsive design */
@media (max-width: 700px) {

  .container {
    flex-direction: column;
  }

  nav a {
    display: block;
    margin: 8px;
  }

  main {
    width: 95%;
  }
}

```

7. CODE EXPLANATION

Code Element / CSS Feature	Explanation
<!DOCTYPE html>	Declares the document as an HTML5 document.
<meta charset>	Defines UTF-8 character encoding.
<meta name="viewport">	Helps the webpage adapt to different screen sizes.
<link>	Connects the HTML document with the CSS stylesheet.
<header>	Creates the top section containing the title and description.
<nav>	Provides navigation links to different sections.
<main>	Contains the main webpage content.
<section>	Divides the webpage into meaningful content areas.
<article>	Groups the service articles for CSS Flexbox layout.
.container	Groups the course cards and is used for Flexbox layout.
<form>	Creates the contact form area.
<input>	Provides fields for entering student information.
<textarea>	Provides a multi-line message input area.
<button type="submit">	Provides the form submit button.
display: flex	Arranges course cards horizontally.
:hover	Changes the appearance of an element when the pointer moves over it.
@media	Applies different CSS rules for smaller screen sizes.
flex-direction: column	Changes the course cards from a row to a column on small screens.
gap	Creates space between the course cards.

8. TEST CASES AND EXECUTION DATA

The checks below cover the visible structure and controls implemented in contact.html and the CSS behavior represented by the Week 01 output.

Test	Test Case	Expected Result	Actual Result	Status
TC-01	Open webpage	Student Learning Portal Actual Result should load successfully.	Webpage loaded successfully.	Pass
TC-02	Check navigation	Home, About, Courses and Contact links should be visible.	All navigation links are displayed.	
TC-03	Check course section	Three course cards should be displayed.	Web Development, Python Programming and Data Science are displayed.	Pass
TC-04	Check contact form	Name, email, message and Send controls should appear.	All form controls are visible.	Pass
TC-05	Check email field	Email input should accept an email-type value.	Email input control is displayed.	Pass
TC-06	Check responsive layout	Course cards should stack vertically on smaller screens.	Cards change to a column layout.	Pass

9. OUTPUT

The screenshot displays a web application interface. At the top, there are three purple-bordered cards with white text, each containing a course description. Below these is a white contact form titled 'Get In Touch' with a purple header. The form includes three input fields: 'Student Name' (with placeholder 'Enter student name'), 'Email Address' (with placeholder 'Enter email address'), and 'Your Message' (a larger text area). A purple 'Send' button is located at the bottom of the form. The footer is a dark grey bar with the text '© 2026 Student Learning Portal'.

Output 1: HTML5 & CSS3 Webpage (reproduced from the repository output layout)

The output displays a styled header, navigation bar, About section, three course cards, contact form, and footer. The webpage also provides responsive behavior for smaller screen sizes.

10. RESULT

Thus, a modern webpage was successfully developed using HTML5 and CSS3. Semantic HTML elements, navigation, content sections, Flexbox, course cards, a contact form, hover effects, and responsive media-query styling were implemented and verified successfully.